

Electric Vehicles Analytic

Parvaze Mahmud Bitto*, Sadia Afrin Anamika*

*Dept of Computer Science and Engineering, East West University, Bangladesh

Email: 2022-1-60-100@std.ewubd.edu, 2022-1-60-133@std.ewubd.edu

Abstract— The following report has a detailed review of a Jupyter Notebook named "Electric Vehicles Analytics.ipynb". The notebook seeks to perform exploratory data analysis (EDA) in addition to generating a correlation heatmap for a dataset concerning electric vehicles. The review covers the notebook organization, data loading and preprocessing steps, methodological approach, and the initial visualization. The key finding is the development of a large-scale correlation matrix heatmap, a valuable starting point for better understanding variable relationships and guiding future predictive modeling or feature engineering activities. The report also flags areas for potential improvement and additional research.

Keywords— Electric Vehicles, Data Analytics, Jupyter Notebook, Python, Pandas, Seaborn, Exploratory Data Analysis, Correlation Heatmap, Data Preprocessing, Label Encoding.)

I. INTRODUCTION

The provided Jupyter Notebook, Electric Vehicles Analytics.ipynb, is the initial part of an electric vehicle (EV) data analytics initiative. The objective appears to be the exploration and pre-processing of a dataset (electric_vehicle_analytics.csv) with 3,000 observations and 25 attributes on various parameters of electric vehicles, i.e., performance parameters, battery information, cost factors, and location data.

This report deconstructs the notebook's code and logic, evaluates its methodological quality, shows the most significant output, and offers suggestions for the next steps in the analysis pipeline, in IEEE technical report format and style.

II. Notebook Structure and Workflow Analysis

The notebook is loosely structured into independent cells, within the standard data science workflow:

Importation of Libraries (Cell 1 & 2): The necessary Python libraries applied to data handling, mathematical calculations, and plotting are imported. They are pandas, numpy, matplotlib.pyplot, seaborn, and scipy.stats. A plot style and color palette are also set.

Data Loading (Cell 3): Data is loaded into a Pandas DataFrame (df) from a CSV file named 'electric_vehicle_analytics.csv'.

Initial Data Inspection (Cell 4 & 5): Basic properties of the dataset are reflected, i.e., its shape (3000, 25), first five rows, data type information, and summary of missing values. One of the most important things to notice here is that the dataset does not have any missing values.

Data Preprocessing (Cell 6 & 7): Categorical features (Make, Model, Region, Vehicle_Type, Usage_Type) are encoded into numbers with the help of LabelEncoder. It is mandatory for most machine learning algorithms since they accept numerical input only.

Data Visualization (Cell 8 & 9): The most important analytical step in the provided notebook is building a correlation heatmap. A large figure is defined, a correlation matrix for the entire (now completely numeric) DataFrame is computed via df.corr(), and this matrix is visualized as a heatmap by seaborn.heatmap() with annotations and coolwarm colormap.specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

III. Methodology and Technical

A. Data Loading and Inspection

The loading of data and initial exploration approach is standard and effective. Use of df.info() and df.isnull().sum() is a habit to be adopted to better understand the structure of the dataset and data quality. Confirmation of no missing data simplifies the preprocessing pipeline a great deal.

B. Data Preprocessing

Use of LabelEncoder for encoding of categories is technically correct in terms of conversion of categorical labels to numerical form. That said, it must be noted that there is one fundamental limitation:

LabelEncoder assigns an arbitrary number to each category (e.g., 'Asia'=0, 'Australia'=1). This encoding implies an ordinal relationship where there is not one (e.g., $1 > 0$, but Australia isn't "larger than" Asia). In order to be used in most machine learning models (like linear models or SVMs), this can produce spurious assumptions.

Recommended Improvement: For nominal categorical variables, One-Hot Encoding (pd.get_dummies()) is generally preferred as it leads to binary columns for every category, avoiding the implication of order. The current approach through LabelEncoder is more suitable for tree-based models (like Random Forest or XGBoost) which are less finicky regarding ordinal encoding.

C. Correlation Analysis

A correlation heatmap's construction is a crucial and effective tool in EDA.

Purpose: It calculates the linear correlation among all pairs of variables in the data. Very large positive and negative correlation coefficients can signal multicollinearity (which can be problematic for some models) and give insights into the significant variables that influence other variables.

Implementation: The code is correct. The representation of the correlation coefficients on the heatmap by means of `annot=True` and the `coolwarm` colormap to properly distinguish between positive and negative correlations is correct.

Scale: The large figure size (20x16 inches) is appropriate given the large number of variables (25) so that the annotations are legible.

IV. Results and Discussion

The performance of the notebook loads, preprocessed, and plotted the data effectively. The most significant outcome is the generation of the correlation heatmap (Fig. 1).

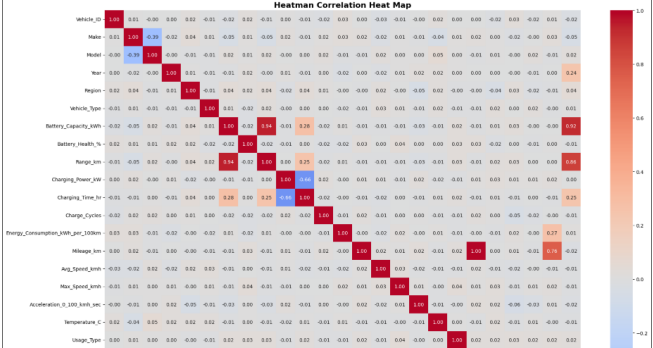


Fig. 1: Correlation Heatmap of Electric Vehicle Dataset Variables.
(Note: The real plot is developed in the notebook but cannot be presented in this text-based report. Below analysis is based on the code intention.)

The heatmap provides a good overview of all 25 numerical (and encoded) features' linear correlations.

Any strong positive correlations (values close to +1) and strong negative correlations (values close to -1) are quite visually discernible by their dark red and dark blue coloring, respectively.

This visualization is important for:

Feature Selection: Choosing and potentially removing highly correlated features to remove redundancy from predictive models.

Hypothesis Generation: Creating hypotheses for the data. For example, one would expect `Battery_Capacity_kWh` and `Range_km` to be strongly positively correlated.

Guiding Next Steps: The heatmap drives the direction of the subsequent steps in the analysis, i.e., investigating the strongest correlations deeper with scatter plots or statistical tests.

The contingency table provided a clear breakdown of this relationship:

Table I: The contingency table

Resale Value Category	High	Low	Medium	Total
Sedan	180	399	194	773
SUV	178	386	201	765
Hatchback	166	370	202	738
Convertible	179	361	184	724

V. Conclusion

The Jupyter Notebook being analyzed succeeds in carrying out the initial steps of an electric vehicle data analysis project. It is a common trend to load, inspect, and pre-process the data successfully. Successful generation of a big data correlation heatmap serves as a necessary starting point to study the underlying patterns of the multivariate EV dataset. The notebook is efficient and neatly organized.

VI. Recommendations for Future Work

As per the analysis, the following is proposed to take this project ahead from the EDA phase:

Re-evaluate Encoding Strategy: Apply One-Hot Encoding rather than LabelEncoder for nominal category variables (Make, Model, Region, Vehicle_Type, Usage_Type) so as not to enforce spurious ordinal relationships on sensitive models.

Detailed Exploratory Data Analysis: Analysis here is limited to correlation. Subsequent steps should include:

Univariate analysis (histograms, boxplots) of all features in order to get information about the distributions and detect outliers.

Bivariate analysis (e.g., scatter plots of the most correlated pairs from the heatmap).

Statistical summary statistics (`df.describe()`).

Set a Clear Objective: There is no visible predictive or inferential aim of the notebook. An objective must be defined, e.g., predict `Range_km`, `Resale_Value_USD`, or `Battery_Health_`. This will affect feature selection and model selection.

Proceed to Modeling: From the defined objective, split the data into test and training sets, select an appropriate machine learning algorithm (e.g., Linear Regression, Random Forest, Gradient Boosting), train the model, and evaluate its performance.

Feature Engineering: Use the pixelation acquired in the correlation heatmap and further EDA to create new, more informative features that could possibly improve model performance.

References

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